

Mentoring Plan

This plan covers every graduate student and postdoctoral researcher supported by this project, at UC Santa Barbara and at Arizona State University, the subrecipient. Mentoring has three components: individual mentoring by the PIs, participation in departmental activities, and participation in institution-level programs. The components apply to both groups, with the differences noted below.

Individual mentoring by the PIs. Every researcher funded by this project will be advised or co-advised by a PI and will join that PI's laboratory. The four PIs have extensive experience in graduate and postdoctoral mentoring and in placing their mentees in high-quality positions in academia and industry. Each PI will provide:

- 1) *Career counseling.* Regular one-on-one meetings on career goals, progress, and strategy, and facilitated introductions to researchers, industry professionals, and alumni. The PIs will advise directly on the academic job search, on what to expect at an interview, and on the first years of an academic career. Postdoctoral researchers will additionally receive guidance on assembling a research statement and a teaching portfolio.
- 2) *Training in the preparation of proposals, publications, and presentations.* Mentorship in writing and submitting papers and rebuttals; training in presentation skills, including slide design and conveying results to non-specialist audiences; and, for senior students and postdoctoral researchers, participation in drafting sections of real proposals and in responding to reviews.
- 3) *Teaching and mentoring skills.* Opportunities to teach class sessions and to mentor junior researchers, including the undergraduates placed with the project through the Early Research Scholars Program at UCSB.
- 4) *Collaboration across disciplines and institutions.* The project is jointly executed by two universities and engages maintainers and operators outside academia, so researchers will collaborate with people from backgrounds and disciplines other than their own as a matter of course, with the PIs providing guidance on doing so effectively.
- 5) *Responsible professional practices.* Because this project discovers exploitable defects in widely deployed software, mentees will be trained in the coordinated vulnerability disclosure process the project follows, in the authorization and rules-of-engagement requirements that govern any assessment, and in the dual-use considerations that attach to offensive security research, alongside the standard responsible-conduct-of-research curriculum at their institution.

Each mentee will maintain an Individual Development Plan, reviewed with their advising PI at least annually and used to set the goals above against a timeline.

Departmental activities. Researchers will be encouraged to attend departmental mentoring events at both institutions, including seminars on writing and presentation, effective teaching techniques, and career development.

Institution-level programs. At UCSB, the Center for Science and Engineering Partnerships (CSEP) runs an intensive Professional Development Series covering basic, communication, and career skills, in workshops of one to two hours offered roughly bi-weekly and led by faculty and practitioners from STEM departments. At ASU, the mentoring of graduate students will take place through the Center for Cybersecurity and Trusted Foundations. The PIs will track participation and discuss it at the annual project meeting.